INDUSTRY DEEP DIVE

US Payments Industry:
Structure, Players & Value Pools

A comprehensive investigation into money movement, market structure, key participants, and emerging trends shaping the future of payments.

$175T+
Annual Payment Volume

5+
Distinct Participant Layers

$500B+
Industry Revenue Pool

3
Transformation Themes

AGENDA

What we'll cover today

An end-to-end view of the US payments ecosystem

01 **Ecosystem Overview**
How money moves — a macro view

02 **Market Sizing & Value Pools**
Revenue distribution across layers

03 **Payment Flows & Networks**
Card rails, ACH, wires & real-time

04 **Key Players & Roles**
Banks, networks, processors & fintechs

05 **Economics & Fee Structures**
Where the money is made

06 **Transformation Themes**
Real-time, embedded & digital wallets

07 **Strategic Implications**
What it means for participants

The US Payments Ecosystem: A Five-Layer Architecture

CONSUMERS & BUSINESSES

Payers & Payees initiating transactions

BANKS & CREDIT UNIONS

Issuing & acquiring financial institutions

PAYMENT NETWORKS

Visa, Mastercard, Amex, Discover, NACHA

PROCESSORS & GATEWAYS

FIS, Fiserv, TSYS, Stripe, Adyen, Braintree

SETTLEMENT & RAILS

ACH, Fed Wire, RTP, FedNow, SWIFT

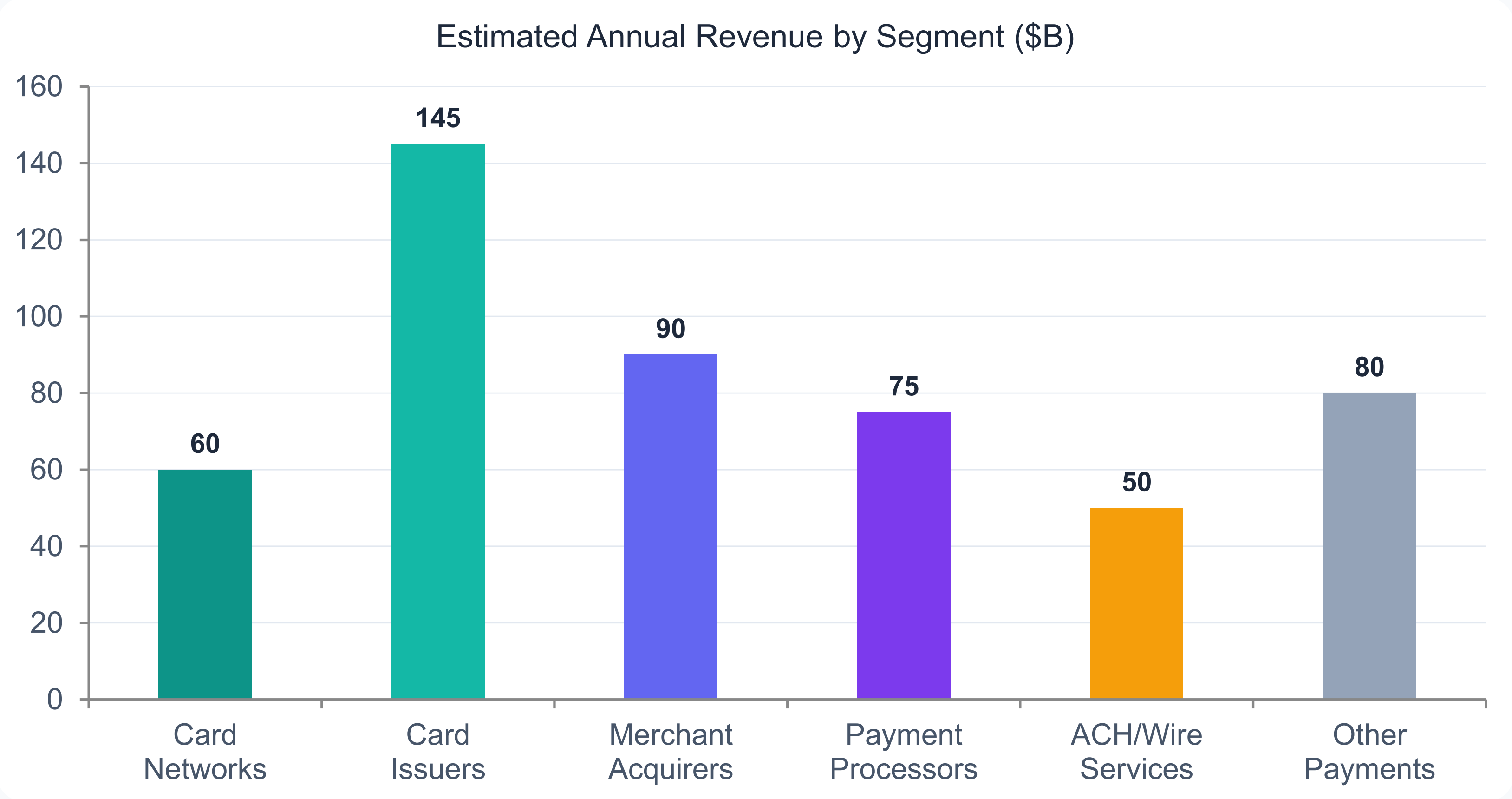
Every payment touches all five layers — generating fees and data at each step

US Payments Revenue Pool: $500B+ Across Segments

$175T
Total payment volume

$500B+
Industry revenue

~1.8%
Blended take rate



Volume by Payment Type

- Consumer Credit Cards 38%
- Consumer Debit Cards 22%
- ACH & Wire Transfers 18%
- Commercial Payments 14%
- Cash & Check 8%

Sources: Federal Reserve, McKinsey Global Payments Report, company filings, analyst estimates

Where Value is Created and Captured in a Card Transaction

Consumer	Merchant	Acquirer	Network	Issuer
Pays $100	Receives ~$97.70	Keeps ~$0.30 (acquiring fee)	Keeps ~$0.15 (network fee)	Keeps ~$1.85 (interchange)

MERCHANT DISCOUNT RATE (MDR) = ~2.3% of transaction

Interchange	Network	Acquiring
~1.85%	~0.15%	~0.30%

KEY INSIGHT:

Interchange — the fee paid by the acquirer's bank to the issuer's bank — represents the largest single fee component (~80%). It is set by networks (Visa/MC) but accrues to issuers, creating misaligned incentives between merchants, networks, and banks.

Durbin Amendment (2010) capped debit interchange at ~$0.21 + 0.05% for large issuers. Premium rewards cards carry higher interchange (1.8–2.5%), subsidizing cardholder benefits at merchant expense.

US Payment Rails: Speed, Cost & Use Cases Compared

Rail	Operator	Settlement Speed	Typical Cost	Annual Volume	Best For
Credit / Debit Card	Visa, MC, Amex	T+1–2 days	1.5%–3.5%	$9T+	Consumer retail
ACH	NACHA / Fed	Same day – T+1	$0.02–$0.50	$80T+	Payroll, B2B, bills
Fedwire	Federal Reserve	Same day (real-time)	$0.82/item	$1,000T+	Large corporate, banks
CHIPS	The Clearing House	Same day (net)	~$0.40/item	$400T+	Interbank, FX
RTP (TCH)	The Clearing House	Real-time (<10 sec)	~$0.045/item	$3T (growing)	P2P, B2B, payroll
FedNow	Federal Reserve	Real-time (<10 sec)	~$0.045/item	Nascent (2023)	Same as RTP, gov't
Checks	Fed Reserve / banks	T+1–2 days	$1–3/item	~$10T (declining)	Legacy B2B, real estate

REAL-TIME

Green text = real-time settlement

Payments Ecosystem Participants: Roles & Relationships

NETWORKS

Set rules, own brand, earn network fees, arbitrate disputes

- Visa
- Mastercard
- American Express
- Discover
- UnionPay

ISSUING BANKS

Issue cards/accounts to consumers, earn interchange & interest

- JPMorgan Chase
- Bank of America
- Citi
- Capital One
- Wells Fargo

ACQUIRING BANKS & PROCESSORS

Onboard merchants, accept card txns, settle funds, earn acquiring fees

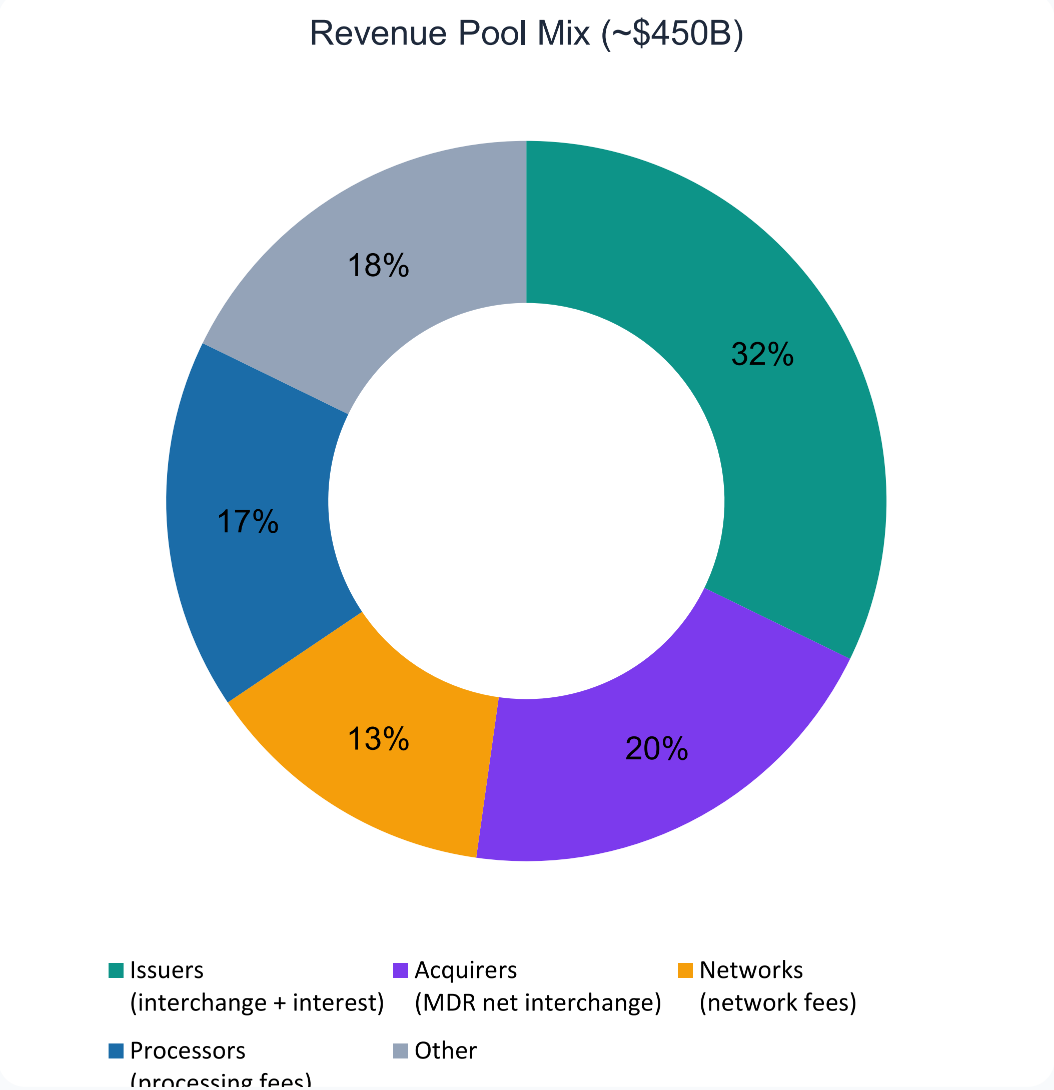
- FIS
- Fiserv
- TSYS
- WorldPay
- Stripe
- Adyen

FINTECHS & DISRUPTORS

Unbundle services, embed payments, create new UX & business models

- PayPal / Venmo
- Square / Block
- Chime
- Klarna
- Affirm
- Plaid

Revenue Model Comparison Across Participant Types



Revenue Model & Margin Profile

Card Networks

~60–65% Op. Margin

Volume-based fees (basis points on GDV); assessment fees; data & value-added services

Card Issuers

~30–35% Op. Margin

Interchange revenue; net interest income on revolving balances; annual card fees; late fees

Merchant Acquirers

~20–30% Op. Margin

MDR spread over interchange; equipment rental; SaaS/software; value-added services

Payment Processors

~20–25% Op. Margin

Per-transaction processing fees; monthly SaaS; implementation & support fees

Three Forces Reshaping the Payments Landscape

01 Real-Time Payments

From batch to instant: a $15T+ addressable opportunity

- FedNow launched July 2023; TCH RTP rail has been live since 2017
- Real-time is becoming table stakes for consumer and B2B payments
- Opportunity to eliminate float, reduce fraud exposure, and accelerate cash flow
- ACH batch volumes ($80T+) are the key migration prize

02 Embedded Finance & BaaS

Payments embedded in every software platform

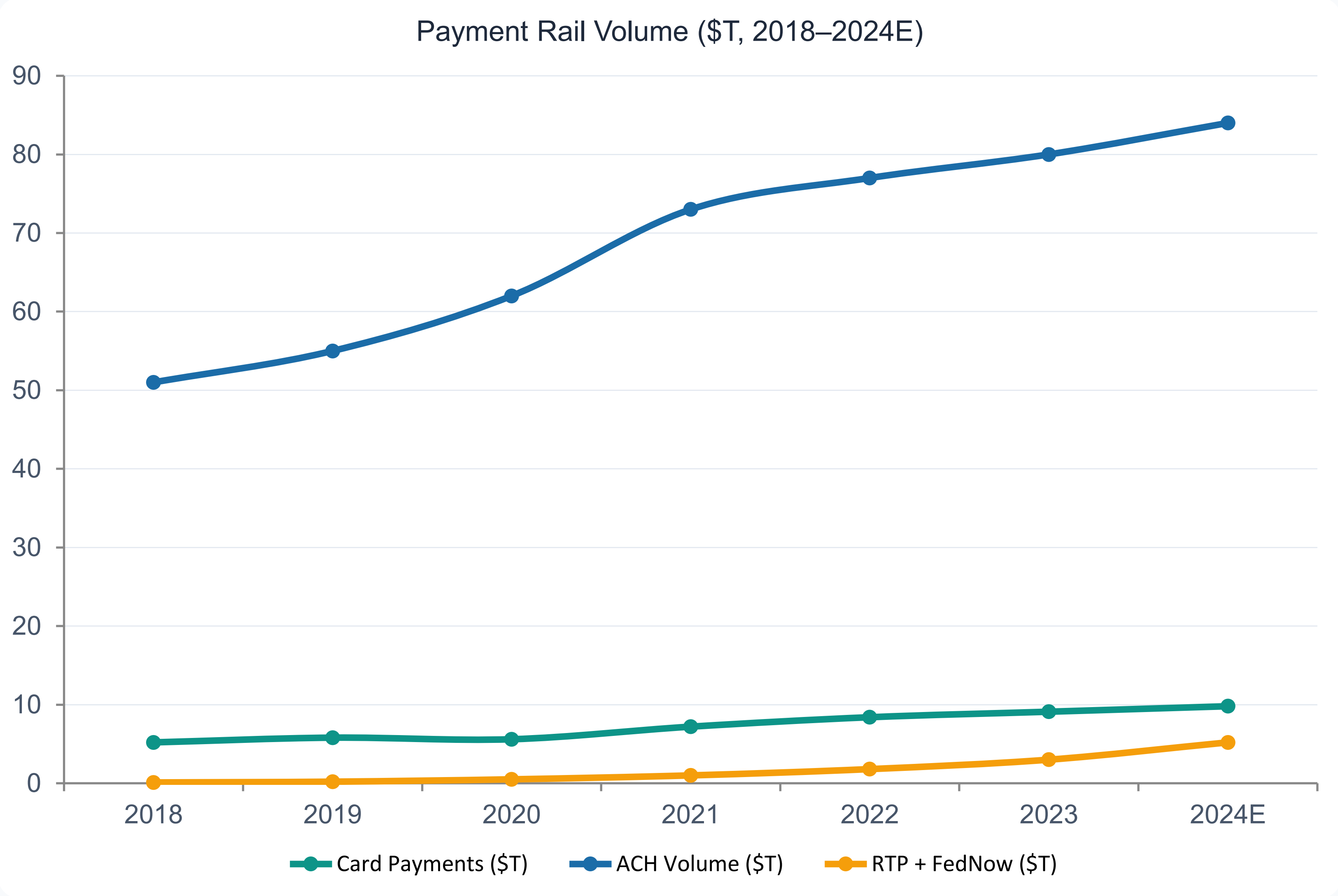
- Banking-as-a-Service (BaaS) players: Unit, Synctera, Treasury Prime, Column
- Software platforms now offering accounts, cards, and lending
- Gross payment volume via embedded channels growing >40% annually
- ISVs and SaaS players taking a meaningful share of acquiring economics

03 Digital Wallets & BNPL

New payment UX displacing traditional card rails

- Apple Pay, Google Pay, PayPal dominate mobile wallet share
- BNPL (Klarna, Affirm, Afterpay) capture 3–5% of e-commerce GMV
- Wallets increasingly tokenize card credentials, reducing fraud losses
- Big Tech encroachment: Apple, Google, Amazon building payments moats

Shifting Mix: Real-Time Rails Are Gaining Fast



Analyst Takeaways

RTP Acceleration

Real-time rails growing at 60–80% CAGR from a small base; FedNow expected to double total real-time capacity.

ACH Durability

ACH remains the backbone of high-value B2B and payroll flows — the platform for most real-time migration.

Card Volume Resilient

Cards grow steadily; e-commerce and contactless adoption offset any cash-to-card plateau effects.

Implications for Each Ecosystem Participant

Banks (Issuer + Acquirer)

⚠ **THREAT**

Interchange regulation; BNPL displacing revolving credit; fintechs with lower-cost structures

✦ **OPPORTUNITY**

Own the real-time rail relationship; build data monetization; white-label BaaS

→ **ACTION**

Invest in real-time infrastructure; partner with or acquire fintechs; rationalize acquiring

Card Networks

⚠ **THREAT**

Account-to-account (A2A) payments bypassing card rails; regulatory scrutiny on interchange

✦ **OPPORTUNITY**

Expand into value-added services (fraud, data, B2B); acquire RTP/fintech assets

→ **ACTION**

Diversify revenue beyond card assessments; compete in A2A via Visa Direct / MC Send

Processors & Fintechs

⚠ **THREAT**

Commoditization of pure processing; margin compression from embedded competitors

✦ **OPPORTUNITY**

Vertical SaaS + payments bundling; international expansion; middleware for RTP

→ **ACTION**

Move up the stack to software; build proprietary distribution; consolidate mid-market

Merchants

⚠ **THREAT**

Rising card acceptance costs; chargeback risk; PCI compliance burden

✦ **OPPORTUNITY**

Real-time A2A payments dramatically cut acceptance costs; surge pricing enabled

→ **ACTION**

Advocate for and accelerate A2A adoption; invest in digital checkout optimization

SUMMARY

Five Things to Know About US Payments

1. Every transaction generates fees at 5+ layers — the system is highly intermediated with significant value pools at each step.
2. Interchange is the largest single fee, flowing to issuers — and it's under increasing pressure from regulation and A2A alternatives.
3. Real-time payments (RTP, FedNow) are the defining structural shift of this decade, threatening ACH batch and card-present economics.
4. Embedded finance is disaggregating the value chain: software platforms now capture acquiring and issuing economics at scale.
5. Big Tech (Apple, Google, Amazon) represents the single largest long-term threat to traditional network and bank payment revenue.